

Rajalakshmi Engineering College

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Batch: 2028
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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 10_MCQ

Attempt : 1
Total Mark : 15
Marks Obtained : 15

Section 1 : MCQ

1. What will be the output of the following code?

```
import java.util.*;
class Main {
    public static void main(String[] args) {
        HashMap<String, String> map = new HashMap<>();
        map.put("A", "Apple");
        map.put("B", "Banana");
        map.put("C", "Cherry");
        map.replace("B", "Blueberry");
        System.out.println(map);
    }
}
```

Answer

{A=Apple, B=Blueberry, C=Cherry}

Status : Correct

Marks : 1/1

2. What will be the output of the following code?

```
import java.util.*;
class Main {
    public static void main(String[] args) {
        HashMap<String, Integer> map = new HashMap<>();
        map.put("X", 10);
        map.put("Y", 20);
        map.put("Z", 30);
        map.remove("Y");
        System.out.println(map);
    }
}
```

Answer

{X=10, Z=30}

Status : Correct

Marks : 1/1

3. What will happen if you add elements in descending order in a TreeSet?

Answer

They are sorted in ascending order

Status : Correct

Marks : 1/1

4. Which of the following is true about TreeMap?

Answer

It maintains natural ordering

Status : Correct

Marks : 1/1

5. What will happen if you add a null element to a TreeSet?

Answer

An exception occurs

Status : Correct

Marks : 1/1

6. Which method removes all elements from a Set?

Answer

clear()

Status : Correct

Marks : 1/1

7. Which of the following allows null keys in Java?

Answer

HashMap

Status : Correct

Marks : 1/1

8. What happens if two keys have the same hash code in a HashMap?

Answer

A linked list is used to store values with the same hash

Status : Correct

Marks : 1/1

9. Which of the following is true about HashMap?

Answer

It is not synchronized

Status : Correct

Marks : 1/1

10. What will be the output of the following code?

```
import java.util.*;
```

```
class Main {  
    public static void main(String[] args) {  
        HashMap<String, Integer> map = new HashMap<>();  
        map.put("A", 1);  
        map.put("B", 2);  
        map.put("C", 3);  
        System.out.println(map.containsKey("B"));  
    }  
}
```

Answer

true

Status : Correct

Marks : 1/1

11. Which statement is true about HashSet and TreeSet?

Answer

TreeSet provides sorted elements

Status : Correct

Marks : 1/1

12. What is the time complexity of retrieving an element from a HashSet?

Answer

O(1)

Status : Correct

Marks : 1/1

13. Which method retrieves the lowest key in a TreeMap?

Answer

firstKey()

Status : Correct

Marks : 1/1

14. What happens when you add duplicate elements to a HashSet?

Answer

The duplicate is ignored

Status : Correct

Marks : 1/1

15. How does HashSet check for duplicate elements?

Answer

Using equals() and hashCode()

Status : Correct

Marks : 1/1

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

A city traffic management system needs to track vehicles entering a toll booth. Each vehicle is uniquely identified by its registration number. The system should allow adding vehicles to a record, ensuring that no duplicate registration numbers exist. The vehicles should be stored in a HashSet, which does not guarantee any specific order.

Your task is to implement a program using a HashSet that allows adding vehicle details and displaying the records.

Input Format

The first line of input contains an integer N - the number of vehicles.

The next N lines contain details of each vehicle in the format: "RegNumber

OwnerName VehicleType"

1. RegNumber (String) - A unique registration number (Alphanumeric).
2. OwnerName (String) - The name of the vehicle owner.
3. VehicleType (String, Car, Bike, or Truck) - The type of vehicle.

If a vehicle with the same registration number is already present, ignore the duplicate entry.

Output Format

The output prints the unique vehicle records in any order (since HashSet does not maintain order).

Output format: "RegNumber OwnerName VehicleType"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

KA01AB1234 John Car

MH02CD5678 Alice Bike

DL03EF9012 Bob Truck

TN04GH3456 Mike Car

KA01AB1234 John Car

Output: TN04GH3456 Mike Car

KA01AB1234 John Car

MH02CD5678 Alice Bike

DL03EF9012 Bob Truck

Answer

// You are using Java

```
import java.util.*;
```

```
class Vehicle {
```

```
    private String regNumber;
```

```
    private String ownerName;
```

```
    private String vehicleType;
```

```
    public Vehicle(String regNumber,String ownerName,String vehicleType) {
```

```
        this.regNumber=regNumber;
```

```
        this.ownerName=ownerName;
```

```

        this.vehicleType=vehicleType;
    }
    public boolean equals(Object obj) {
        if (this == obj)
            return true;
        if (obj == null||getClass()!=obj.getClass())
            return false;
        Vehicle vehicle = (Vehicle) obj;
        return regNumber.equals(vehicle.regNumber);
    }
    public int hashCode() {
        return Objects.hash(regNumber);
    }
    public String toString() {
        return regNumber + " " + ownerName + " " + vehicleType;
    }
}
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = Integer.parseInt(sc.nextLine());
        HashSet<Vehicle> vehicleSet = new HashSet<>();
        for (int i = 0; i < N; i++) {
            String regNumber = sc.next();
            String ownerName = sc.next();
            String vehicleType = sc.next();
            Vehicle v = new Vehicle(regNumber,ownerName,vehicleType);
            vehicleSet.add(v);
        }
        for (Vehicle v:vehicleSet) {
            System.out.println(v);
        }
    }
}

```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

John is organizing a fruit festival, and the quantities of various fruits are stored in a HashMap where fruit names are keys and quantities are values.

Help him develop a program to find the total quantity of fruits for the festival by summing up the values in the HashMap.

Input Format

The input consists of fruit quantities in the format 'fruitName:quantity', where fruitName is the name of the fruit(a string), and quantity is a double value representing the quantity.

The input is terminated by entering "done".

Output Format

The output prints a double value, representing the sum of values in the HashMap, rounded off to two decimal places.

If the value is not numeric, print "Invalid input".

If any special characters other than ':' are entered, print "Invalid format".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: Banana:15.2

Orange:56.3

Mango:47.3

done

Output: 118.80

Answer

// You are using Java

import java.util.*;

class FruitFestival {

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

HashMap<String, Double> fruits = new HashMap<>();

double total = 0.0;

boolean invalidInput = false;

boolean invalidFormat = false;

while (true) {

String input = sc.nextLine().trim();

if (input.equalsIgnoreCase("done")) {
break;

}

if (!input.contains(":")) {
invalidFormat = true;
break;

}

String[] parts = input.split(":");

if (parts.length != 2) {
invalidFormat = true;

```

        break;
    }
    String fruitName = parts[0].trim();
    String quantityStr = parts[1].trim();
    double quantity;
    try {
        quantity = Double.parseDouble(quantityStr);
    } catch (NumberFormatException e) {
        invalidInput = true;
        break;
    }
    if (quantity < 1.0 || quantity > 100.0) {
        invalidInput = true;
        break;
    }

    fruits.put(fruitName, quantity);
}
sc.close();

if (invalidFormat) {
    System.out.println("Invalid format");
} else if (invalidInput) {
    System.out.println("Invalid input");
} else {
    for (double value : fruits.values()) {
        total += value;
    }
    System.out.printf("%.2f\n", total);
}
}
}

```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

Priya is analyzing encrypted messages in a research project. She wants to analyze the frequency of each character in a given paragraph. The characters should be stored in a TreeMap so that the output is sorted in ascending order of characters automatically.

You are required to build a Java program that:

Uses a `TreeMap<Character, Integer>` to count how many times each character appears in the message. Ignores spaces and considers only alphabets (case-sensitive). Outputs the frequencies of characters in sorted order.

You must use a TreeMap in the class named MessageAnalyzer.

Input Format

The first line of input contains an integer n, the number of lines in the message.

The next n lines each contain a string (the encrypted message line).

Output Format

The first line of output prints: "Character Frequency:"

Then print each character and its frequency in the format: "<character>: <count>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2
Hello World
Java

Output: Character Frequency:

H: 1

J: 1

W: 1

a: 2

d: 1

e: 1

l: 3

o: 2

r: 1

v: 1

Answer

```
// You are using Java
import java.util.*;
class MessageAnalyzer {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        TreeMap<Character, Integer> frequencyMap = new TreeMap<>();
        int n = Integer.parseInt(sc.nextLine().trim());
        for (int i = 0; i < n; i++) {
            String line = sc.nextLine();
            for (char ch : line.toCharArray()) {
                if (Character.isLetter(ch)) {
```

```
        frequencyMap.put(ch, frequencyMap.getOrDefault(ch, 0) + 1);
    }
}
System.out.println("Character Frequency:");
for (Map.Entry<Character, Integer> entry : frequencyMap.entrySet()) {
    System.out.print(entry.getKey() + ": " + entry.getValue() + " ");
}
System.out.println();
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

In a ticket reservation system, you store the available seat numbers in a TreeSet. Users input their desired seat number, and the program checks whether the chosen seat is available.

Using a TreeSet ensures quick and efficient verification of seat availability, ensuring a smooth and organized ticket booking process.

Input Format

The first line of input contains a single integer n , representing the number of available seats.

The second line contains n space-separated integers, representing the available seat numbers.

The third line contains an integer m, representing the seat number that needs to be searched.

Output Format

The output displays "[m] is present!" if the given seat is available. Otherwise, it displays "[m] is not present!"

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 4

2 4 5 6

5

Output: 5 is present!

Answer

// You are using Java

```
import java.util.*;
```

```
class Main{
```

```
    public static void main(String[] args){
```

```
        Scanner sc=new Scanner(System.in);
```

```
        int n=sc.nextInt();
```

```
        Set<Integer>seats=new TreeSet<>();
```

```
        for(int i=0;i<n;i++){
```

```
            seats.add(sc.nextInt());
```

```
        }
```

```
        int m=sc.nextInt();
```

```
        if(seats.contains(m)){
```

```
            System.out.println(m+" is present!");
```

```
        }
```

```
        else{
```

```
            System.out.println(m+" is not present!");
```

```
        }
```

```
    }
```

```
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 10_PAH

Attempt : 1
Total Mark : 30
Marks Obtained : 30

Section 1 : Coding

1. Problem Statement

Sarah is working on a spam detection system that analyzes incoming messages for unique patterns. Spammers often use repetitive character sequences, making it important to identify the first non-repeating character in a message.

Given a string, Sarah needs to determine the first character that appears only once. If all characters repeat, the system should return -1.

She decides to use a HashMap to efficiently track character frequencies and find the solution.

Input Format

The first line contains an integer N representing , the length of the string.

The second line contains a string of N lowercase English letters (a-z).

Output Format

The output prints a character representing the first non-repeating character. If none exist, print -1.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 10
abacabadac

Output: d

Answer

```
// You are using Java
import java.util.*;
class SpamDetector {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = Integer.parseInt(sc.nextLine().trim());
        String str = sc.nextLine().trim();
        HashMap<Character, Integer> charCount = new HashMap<>();
        for (char ch : str.toCharArray()) {
            charCount.put(ch, charCount.getOrDefault(ch, 0) + 1);
        }
        char result = '-';
        for (char ch : str.toCharArray()) {
            if (charCount.get(ch) == 1) {
                result = ch;
                break;
            }
        }
        if (result == '-') {
            System.out.println("-1");
        } else {
            System.out.println(result);
        }
    }
}
```

```
        sc.close();  
    }  
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

A university maintains a list of student records and wants to store them in a sorted manner based on their GPA. If two students have the same GPA, they should be further sorted by their name in lexicographical order. Implement a program that uses a TreeSet to store student records and ensures unique student IDs.

Input Format

The first line contains an integer N - the number of students.

The next N lines contain details of each student in the format: "StudentID Name GPA"

- StudentID (Integer) - A unique identifier.
- Name (String) - The student's name (can contain spaces).
- GPA (Double) - The Grade Point Average.

Output Format

The output prints the list of students in ascending order of GPA.

If two students have the same GPA, sort them by name.

Print details in the format: "StudentID Name GPA" in the output, GPA is rounded to two decimal places.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5
101 John 8.5

102 Alice 9.1
103 Bob 8.5
104 Zoe 7.3
105 Charlie 9.1

Output: 104 Zoe 7.30
103 Bob 8.50
101 John 8.50
102 Alice 9.10
105 Charlie 9.10

Answer

// You are using Java

```
import java.util.*;
```

```
import java.text.DecimalFormat;
```

```
class Student implements Comparable<Student> {
```

```
    private int id;
```

```
    private String name;
```

```
    private double gpa;
```

```
    public Student(int id, String name, double gpa) {
```

```
        this.id = id;
```

```
        this.name = name;
```

```
        this.gpa = gpa;
```

```
    }
```

```
    public int getId() {
```

```
        return id;
```

```
    }
```

```
    public String getName() {
```

```
        return name;
```

```
    }
```

```
    public double getGpa() {
```

```
        return gpa;
```

```
    }
```

```
    @Override
```

```
    public int compareTo(Student other) {
```

```
        if (Double.compare(this.gpa, other.gpa) != 0) {
```

```
            return Double.compare(this.gpa, other.gpa);
```

```
        } else {
```

```
            int nameCompare = this.name.compareTo(other.name);
```

```
            if (nameCompare != 0) {
```

```
                return nameCompare;
```

```
            } else {
```

```
                return Integer.compare(this.id, other.id);
```

```

    }
    }
    }
}

class StudentSorter {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        TreeSet<Student> students = new TreeSet<>();
        HashSet<Integer> ids = new HashSet<>();
        for (int i = 0; i < n; i++) {
            int id = sc.nextInt();
            String name = sc.next();
            double gpa = sc.nextDouble();
            if (!ids.contains(id)) {
                ids.add(id);
                students.add(new Student(id, name, gpa));
            }
        }
        sc.close();
        DecimalFormat df = new DecimalFormat("0.00");
        for (Student s : students) {
            System.out.println(s.getId() + " " + s.getName() + " " +
df.format(s.getGpa()));
        }
    }
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Riya is building a calendar event scheduler where each event is stored in chronological order using a TreeMap. The key represents the event time in 24-hour format (HH:MM), and the value is the event description.

She wants the system to:

Automatically sort events by time. Avoid duplicate time entries — if a duplicate time is entered, ignore the new entry. Print all scheduled events in order.

Implement this logic using a class named EventManager.

Input Format

The first line of the input contains an integer n, representing the number of events.

The next n lines each contain a string in the format: "HH:MM Description"

(Example: 09:00 TeamMeeting).

Output Format

The first line of the output prints "Scheduled Events:"

The next k lines print each event in the format: "HH:MM - Description"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

09:00 TeamMeeting

13:30 LunchBreak

11:00 ProjectUpdate

09:00 Standup

15:00 ClientCall

Output: Scheduled Events:

09:00 - TeamMeeting

11:00 - ProjectUpdate

13:30 - LunchBreak

15:00 - ClientCall

Answer

// You are using Java

```
import java.util.*;
```

```
class EventManager {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        TreeMap<String, String> events = new TreeMap<>();
```

```
        int n = sc.nextInt();
```

```
for (int i = 0; i < n; i++) {  
    String time = sc.next();  
    String description = sc.next();  
    if (!events.containsKey(time)) {  
        events.put(time, description);  
    }  
}  
sc.close();  
System.out.println("Scheduled Events:");  
for (Map.Entry<String, String> entry : events.entrySet()) {  
    System.out.println(entry.getKey() + " - " + entry.getValue());  
}  
}  
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 10_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : COD

1. Problem Statement

David is managing an employee database where each employee has a unique ID, name, and department. He wants to ensure that duplicate employee IDs are not added to the system. Implement a Java program that allows adding employees to the system, displaying all employees, and checking if an employee exists based on the given ID.

Implement a class EmployeeDatabase that contains a HashSet to store employee records. The Employee class should be a user-defined object containing employee details. The main class should handle user operations and interact with the EmployeeDatabase class.

Input Format

The first line contains an integer n representing the number of employees to be added.

The next n lines follow, each containing:

1. An integer employee_id
2. A string name
3. A string department

The next line contains an integer m representing the number of queries.

The next m lines follow, each containing an employee ID to check for existence.

Output Format

The output prints a list of all employees added in the format:

"ID: <employee_id>, Name: <name>, Department: <department>"

For each query, output "Employee exists" if the ID is found, otherwise "Employee not found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

101 John IT

102 Alice HR

103 Bob Finance

2

101

104

Output: ID: 101, Name: John, Department: IT

ID: 102, Name: Alice, Department: HR

ID: 103, Name: Bob, Department: Finance

Employee exists

Employee not found

Answer

```
import java.util.*;
```

```
class Employee {
```

```
private int id;
private String name;
private String department;

public Employee(int id, String name, String department) {
    this.id = id;
    this.name = name;
    this.department = department;
}

public int getId() {
    return id;
}

@Override
public boolean equals(Object obj) {
    if (this == obj) return true;
    if (obj == null || getClass() != obj.getClass()) return false;
    Employee emp = (Employee) obj;
    return id == emp.id;
}

@Override
public int hashCode() {
    return Objects.hash(id);
}

@Override
public String toString() {
    return "ID: " + id + ", Name: " + name + ", Department: " + department;
}
}

class EmployeeDatabase {
    private HashSet<Employee> employees = new HashSet<>();
    public void addEmployee(int id, String name, String department) {
        employees.add(new Employee(id, name, department));
    }
    public void displayEmployees() {
        for (Employee e : employees) {
            System.out.println(e);
        }
    }
}
```

```

    public boolean checkEmployee(int id) {
        for (Employee e : employees) {
            if (e.getId() == id) return true;
        }
        return false;
    }
}

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        EmployeeDatabase db = new EmployeeDatabase();
        int n = sc.nextInt();
        for (int i = 0; i < n; i++) {
            int id = sc.nextInt();
            String name = sc.next();
            String department = sc.next();
            db.addEmployee(id, name, department);
        }
        db.displayEmployees();
        int m = sc.nextInt();
        for (int i = 0; i < m; i++) {
            int id = sc.nextInt();
            if (db.checkEmployee(id))
                System.out.println("Employee exists");
            else
                System.out.println("Employee not found");
        }
        sc.close();
    }
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Arjun is working on a program that checks if one set of numbers is a subset of another. If Set B is a subset of Set A, the program should print "YES" followed by the sorted elements of Set B. If Set B is not a subset of Set A, the program should print "NO" followed by the average of all elements from both sets combined, rounded to two decimal places.

Implement a class Solution with the required method to perform the subset check using TreeSet in Java.

Input Format

The first line contains an integer n - the number of elements in Set A.

The second line contains n space-separated integers - the elements of Set A.

The third line contains an integer m - the number of elements in Set B.

The fourth line contains m space-separated integers - the elements of Set B.

Output Format

If Set B is a subset of Set A, print "YES" followed by the sorted values of Set B.

Otherwise, print "NO" followed by the average of all numbers in both sets (rounded to two decimal places).

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

1 2 3 4 5

3

2 3 5

Output: YES 2 3 5

Answer

```
import java.util.*;
```

```
// You are using Java
```

```
class Solution {
```

```
    public static void checkSubset(TreeSet<Integer> setA, TreeSet<Integer> setB,  
    int totalCount, long sum) {
```

```
        if (setA.containsAll(setB)) {
```

```
            System.out.print("YES");
```

```
            for (int num : setB) {
```

```
                System.out.print(" " + num);
```

```

    }
    System.out.println();
} else {
    double avg = (double) sum / totalCount;
    System.out.printf("NO %.2f\n", avg);
}
}
}

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        TreeSet<Integer> setA = new TreeSet<>();
        long sum = 0;
        for (int i = 0; i < n; i++) {
            int num = sc.nextInt();
            setA.add(num);
            sum += num;
        }
        int m = sc.nextInt();
        TreeSet<Integer> setB = new TreeSet<>();
        for (int i = 0; i < m; i++) {
            int num = sc.nextInt();
            setB.add(num);
            sum += num;
        }
        Solution.checkSubset(setA, setB, n + m, sum);
        sc.close();
    }
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Tony is an e-learning platform administrator, he oversees the user ratings for various online courses offered in the platform.

To enhance user experience, you should assist him in utilizing a HashMap to store course ratings given by learners. Regularly, he analyzes this data to

identify the highest and lowest-rated courses, enabling targeted improvements and ensuring the quality of the educational content. This process assists in maintaining a competitive and engaging online learning environment for the users.

Input Format

The input consists of a string representing the course name followed by a double value representing the course's rating, in separate lines.

The input is terminated by entering "done".

Output Format

The first line of output prints the string "Highest Rated Course: " followed by the highest-rated course.

The second line prints the string "Lowest Rated Course: " followed by the lowest-rated courses.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: DSA

4.0

OOPS

4.2

C

3.2

done

Output: Highest Rated Course: OOPS

Lowest Rated Course: C

Answer

```
import java.util.HashMap;  
import java.util.Map;  
import java.util.Scanner;
```

```
// You are using Java  
class CourseAnalyzer {
```

```

    public Map<String, String>
    identifyHighestAndLowestRatedCourses(Map<String, Double> courseRatings) {
        String highestCourse = "";
        String lowestCourse = "";
        double highestRating = Double.MIN_VALUE;
        double lowestRating = Double.MAX_VALUE;
        for (Map.Entry<String, Double> entry : courseRatings.entrySet()) {
            String course = entry.getKey();
            double rating = entry.getValue();
            if (rating > highestRating) {
                highestRating = rating;
                highestCourse = course;
            }
            if (rating < lowestRating) {
                lowestRating = rating;
                lowestCourse = course;
            }
        }
        Map<String, String> result = new HashMap<>();
        result.put("highest", highestCourse);
        result.put("lowest", lowestCourse);
        return result;
    }
}

```

```

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        Map<String, Double> courseRatings = new HashMap<>();

        while (true) {
            String courseName = scanner.nextLine();
            if (courseName.equalsIgnoreCase("done")) {
                break;
            }
            double rating = Double.parseDouble(scanner.nextLine().trim());
            courseRatings.put(courseName, rating);
        }

        CourseAnalyzer analyzer = new CourseAnalyzer();
        Map<String, String> result =
        analyzer.identifyHighestAndLowestRatedCourses(courseRatings);
    }
}

```

```
System.out.printf("Highest Rated Course: %s\n", result.get("highest"));
System.out.printf("Lowest Rated Course: %s", result.get("lowest"));

    scanner.close();
}
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

A college professor wants to keep track of students who attend classes. Each student has a unique roll number and their attendance count increases every time they attend a class. The system should allow adding a student, marking their attendance, and displaying all students with their total attendance.

Your task is to implement a Java program using TreeSet to maintain students in sorted order of roll numbers and track their attendance count.

Operations:

A roll_no name Add a student with roll number and name (if not already added). M roll_no Mark attendance for the student with the given roll number (increase their count by 1). D Display all students in ascending order of roll number along with their attendance count.

Input Format

The first line contains an integer N - the number of students.

The next N lines contain one of the following commands:

A roll_no name

M roll_no

D

- A (Add) Adds a new student with a unique roll number and name.
- M (Mark) Increases attendance count for the given roll number.
- D (Display) Prints all students in ascending order of roll number.

Output Format

For D, output prints each student's roll number, name, and attendance count in ascending order of roll number.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

A 101 Alice

A 102 Bob

M 101

M 101

D

Output: 101 Alice 2

102 Bob 0

Answer

// You are using Java

import java.util.*;

class Student implements Comparable<Student> {

int rollNo;

String name;

int attendance;

public Student(int rollNo, String name) {

 this.rollNo = rollNo;

 this.name = name;

 this.attendance = 0;

}

@Override

public int compareTo(Student other) {

 return Integer.compare(this.rollNo, other.rollNo);

}

@Override

public boolean equals(Object obj) {

 if (this == obj) return true;

 if (!(obj instanceof Student)) return false;

 Student other = (Student) obj;

 return this.rollNo == other.rollNo;

```

    }
    @Override
    public int hashCode() {
        return Objects.hash(rollNo);
    }
}

class AttendanceSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = Integer.parseInt(sc.nextLine());
        TreeSet<Student> students = new TreeSet<>();
        for (int i = 0; i < N; i++) {
            String input = sc.nextLine();
            String[] parts = input.split(" ");
            switch (parts[0]) {
                case "A":
                    int rollNo = Integer.parseInt(parts[1]);
                    String name = parts[2];
                    Student newStudent = new Student(rollNo, name);
                    students.add(newStudent);
                    break;
                case "M":
                    int markRollNo = Integer.parseInt(parts[1]);
                    for (Student s : students) {
                        if (s.rollNo == markRollNo) {
                            s.attendance++;
                            break;
                        }
                    }
                    break;
                case "D":
                    for (Student s : students) {
                        System.out.println(s.rollNo + " " + s.name + " " + s.attendance);
                    }
                    break;
            }
        }
        sc.close();
    }
}

```

Status : Correct

Marks : 10/10